

(3)

Suitable

Transmeters to 330KV for transmission over long distances to the load centres. Power transmitted or evacuated by a transmission line or network is directly proportional to the square of the voltage, therefore increasing the level of the transmission voltage increases the power transmission capability of the line. Thus the higher the voltage level the greater the line capacity to transmit power. The transmission is then stepped down at the large or bulk power stations where the voltage is reduced from 330KV to 132KV. At this level bulk power industries are being fed with electric energy (Steel mills, Refineries and Petrochemicals, Cement factories, Sugar, beverages etc.). The next step-down is at the secondary transmission stage also known as the subtransmission level where is stepped down from 132KV to 33KV. At this substation power is fed to light industrial loads such as paper mills, canning factories, flour mills etc. The next level is at the distribution level where voltage is from 33KV to 11KV secondary feeders for onward to secondary distribution transformers that supply power small consumers (domestic / household loads, Commercial / Business centres, street lighting etc.).

GRID

(C)

CAUSES OF NATIONAL FAILURE

The electrical power network or grid is exposed to harsh weather conditions, due to these and other factors the network is often affected due to faults and various forms of malfunction that can lead to partial or total network failure or grid collapse. Grid failure occurs due to removal of a large load or fault when a major generator unit is removed from supplying the network due to fault or an out of service removal from services.

(4)

Causes of grid failure can be grouped into three main categories:

(i) Normal Causes:- These fault that arises due to natural circumstances such as accidents, collision or fire, weather induced natural disasters such as floods, land slides, heat waves, earthquakes and heavy storms.

(ii) Attacks:- Human attack such Cyber attack due to failure in Cyber security, physical human attack such as Vandalism and terrorism or theft.

(iii) Operational Errors:- These failures which are related to human errors, equipment malfunctions, ageing and other technical errors.

(I) PRINCIPLES OF ELECTRICAL GENERATION

The Conventional Power generation methods are:

The thermal Power plants (Coal and Nuclear) Plants and the hydro power generation system.

This section considers the principles of operation of the conventional power generation systems.

1. THE HYDRO POWER GENERATION

The hydroelectric power generation method utilises the potential energy of water - The potential energy is the energy stored in a substance

due to its position or state. Water stored behind a dam exerts some form of force when allowed to fall freely due to gravity

over a given distance (head) thereby doing some useful work. In hydro power plant the force exerted by the moving water is harnessed to drive a water turbine or hydraulic turbine.

(5)

This hydraulic turbine is connected to the shaft of an alternator (a synchronous generator) that will generate electricity. Because gravity is involved in producing the force which the water fall the energy stored in the water is called gravitational potential energy. It is the oldest and cheapest power generation method. Figure 2 shows the Siphon block diagram a hydro power plant indicating its major component parts.

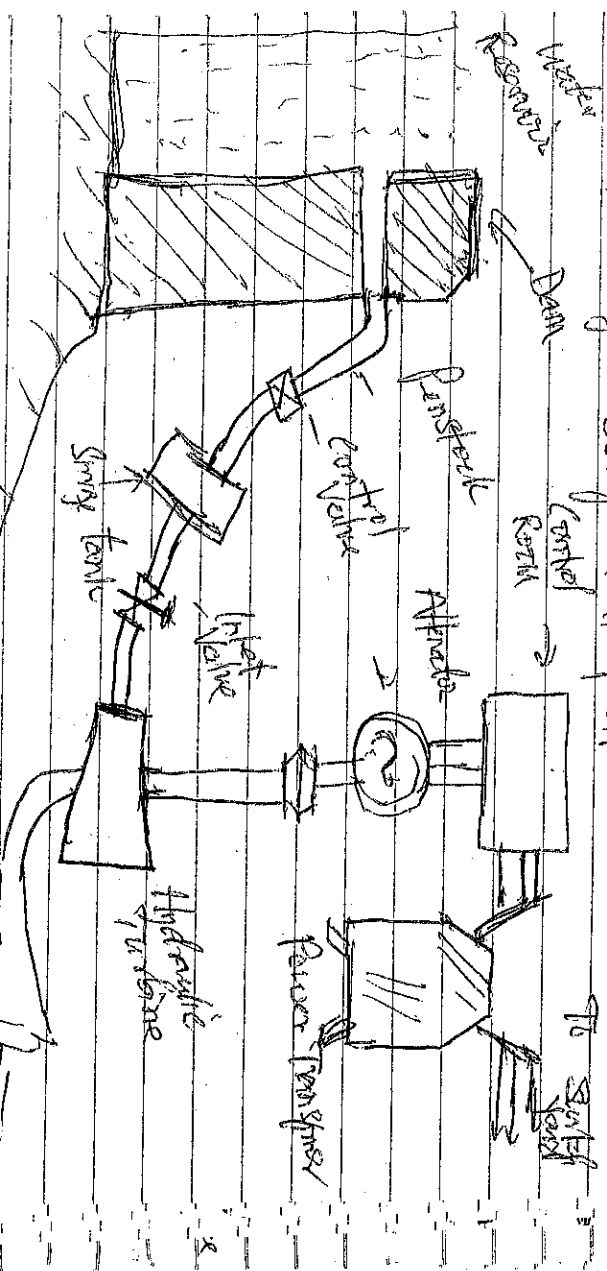


Figure 2: Basic layout of a hydroelectric power plant.

① The Dam:- This is usually a concrete structure, it develops a reservoir to store water upto the required head for power generation i.e. the difference between the upper reservoir level and the tail race.

② Forebay Reservoir:- This is the body of water above the intake. It receives the water released from the catchment area. The catchment area gets the water from the rain, flowing rivers and streams.

(6)

(3) Control gate :- This gate is used to regulate the flow of water from the dam. The intake valve contains trash rack made of steel to filter out debris, leaves and other forms of suspension which may damage the turbine.

(4) The penstock :- This is a closed conduit ~~for~~ pipe for supplying water under pressure from storage to the turbines. It is made of steel for low discharge and reinforced concrete for high pressure discharge.

(5) Surge Tank :- The surge tank is located near the turbine as a small reservoir. It is used to reduce the impact of rapid pressure variations which lead to damage if transmitted through the penstock. The water channel has to withstand the sudden changes in water due to increase or decrease in load demand. This is referred to as "water hammer".

(6) Tail Race :- This is the channel that carries water away from the turbine and discharge it into the main river.

FACTORS AFFECTING SITE SELECTION OF A HYDRO POWER PLANT (HPP).

The following factors are to be considered in selecting the site for construction of a hydro power plant:

(1) Availability of water :- The availability of the hydrological information of the site on precipitation, evaporation and surface water/run off water data for several years is to be ascertained and obtained.

(2) Storage capacity :- Due to fluctuation in the flow of rivers/streams, storage capacity

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Must be ascertained from hydrograph of the Site.

③ Sedimentation :- The storage capacity of the dam/reservoir is reduced due to the gradual deposition of silt. This silt may gradually damage the turbines blades.

④ Water Pollution :- Polluted water can damage metallic structures thereby ~~lead~~ leading to the massive operation of the power plant. A good quality water devoid of salts and other harmful substances is an essential factor in site selection for a hydro power plant.

⑤ Environmental Considerations :-

The hydro power project may cause exposure of land area. This can lead to submerging of vast land including villages thereby resulting in displacement of people from their original settlements and means of livelihood. The site selected should ensure safe surroundings, avoid health hazards, and unnecessary displacement.

⑥ Access to :- These power project involves the road/rail line movement & heavy equipment and machinery loads for the civil engineering construction work and the power equipment to be installed. A suitable location accessible to a standard means of transportation is therefore mandatory.

(8) Advantages & Disadvantages of Hydro power plants :-

Advantages :-

- (i) Low Maintenance and operating cost. The efficiency of the plant does not change with years of operation or age.
- (ii) Water is a renewable energy resource.
- (iii) Hydro power plant is a multipurpose project such as river flood control, irrigation, source of drinking water, fishing and navigation.

(i) Hydro power plant has longer life span (100-125 years) as they operate at atmospheric temperature.

(vi) Since hydro power plants runs at low speed (100-400 rpm) this eliminates the need for special steel alloys for the design of the mechanical structures.

(vii) It is completely free from pollution and noise.

Disadvantages :-

Hydro

(1) The initial cost of a power plant is very high.

(2) Since there are usually breasted far away from main load centres, cost of transmission lines will be high.

(3) During drought periods power production will reduce or even stopped due to insufficient water in the reservoir.

(4) Water in the reservoir is lost by evaporation, sedimentation and silt.

(5) It requires long gestation period compared to steam power plants.

(9)

COMPUTATION OF ELECTRICAL POWER GENERATED BY A HYDRO POWER PLANT.

The Power generated by a hydro power plant is given by:

$$P = \rho \cdot 81 \cdot Q \cdot H \quad (\text{KW})$$

ρ water

Q = water discharge rate
(m^3/second)

H = water head or height

η = Fall of water (m)

η = Combined efficiency of
Turbine and generator
systems.

Example 1

A hydro power station has an average water discharge in a year as given in table below.

Table 1

Month	Monthly Discharge Rate (m^3/sec)
January	200
February	400
March	600
April	8400
May	1200
June	1800
July	1600
August	1200
September	2000
October	1200
November	800
December	400

Calculate the average water discharge rate and the Power generated when the head of water is 50m and the combined efficiency of the power station is 90%.
Draw the hydrographs of the site and estimate the total annual storage capacity of the reservoir.

(10)

(a) Total water discharge from the reservoir in a year = Total monthly discharge
 $= 300 + 400 + 2400 + 2200 + 1800 + 1600 + 1200 + 2000 + 800 + 400$
 $= 13,800 \text{ m}^3/\text{Sec.}$

Average Monthly discharge for the year
 $= \frac{\text{Total discharge}}{\text{Number of Months}} = \frac{13,800}{12}$

$= 1150 \text{ m}^3/\text{Sec}$

$\therefore Q = 1150 \text{ m}^3/\text{Sec}$

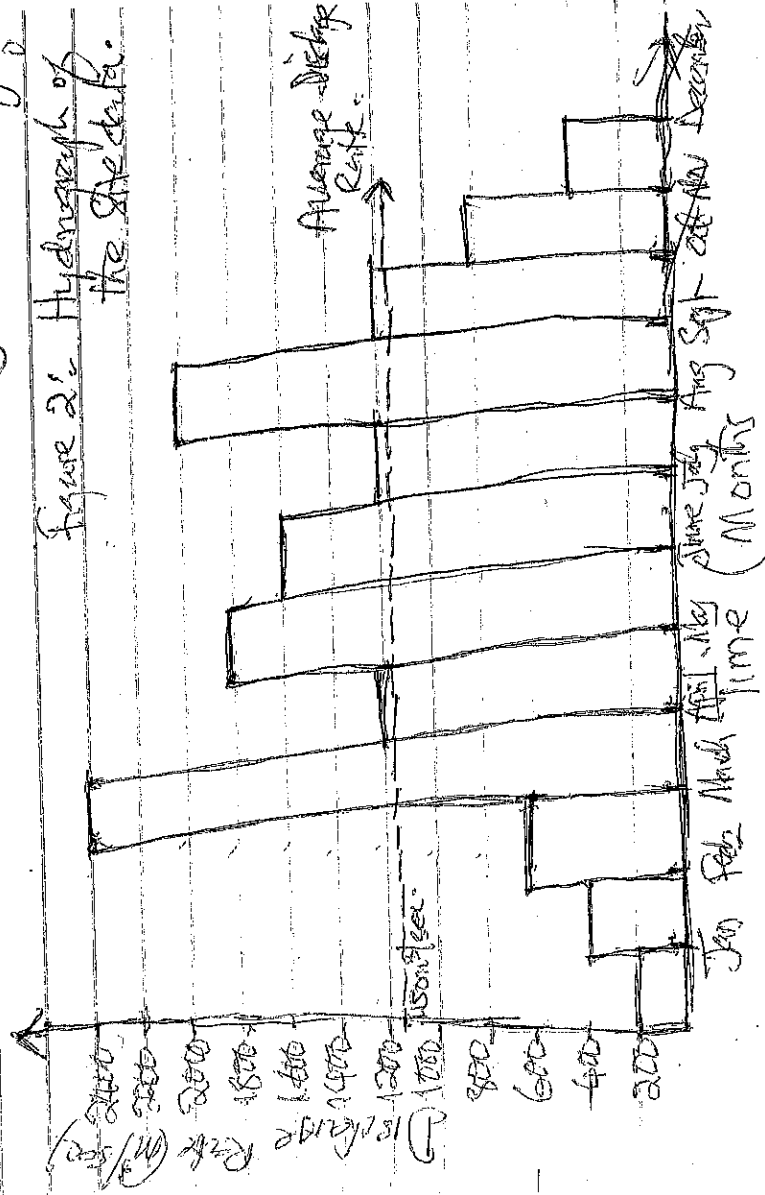
(b) Power generated by the plant

$P = Q \cdot \rho \cdot g \cdot H$

$= 9.81 \times 1150 \times 50 \times 0.9$

$= 507.7 \text{ MW}$

(c) Hydrograph: A graph flow variation against time over a given duration.



(d) Water storage capacity of the reservoir
 Since water is released in the reservoir
 only when the quantity available exceeds
 the discharge rate monthly.

Total storage from the hydrograph:

$$\begin{aligned} \text{Total storage} &= (2400 - 1150) + (2400 - 1150) + (1800 - 1150) + (1500 - 1150) \\ &\quad + (1200 - 1150) + (2000 - 1150) + (1200 - 1150) \\ &= 1250 + 50 + 650 + 450 + 50 + 850 \\ &\quad + 50 = 2350 \text{ m}^3/\text{sec} \end{aligned}$$

$$\text{Monthly discharge} = 3350 \times 30 \times 24 \text{ hrs} \times 60 \text{ m/sec} \times 60 \text{ sec}$$

$$= 8.7 \times 10^9 \text{ m}^3/\text{month}$$

$$\begin{aligned} \text{Annual storage capacity of the} &= 8.7 \times 10^9 \times 12 \text{ months} \\ \text{Reservoir} &= 1.04 \times 10^{11} \text{ m}^3/\text{year} \end{aligned}$$

Thermal Power Generation Station

The chemical energy stored in Coal is transformed into electrical energy in a thermal power station. The heat released by the combustion of coal or oil / natural gas produces steam in a boiler at high temperature and pressure which when passed through a steam turbine give off some of its internal energy as mechanical energy. The thermal power plants are low efficiency plants with a maximum conversion of 40% due to heat losses in combustion gases and large quantity of rejected heat at the condenser.

As the generating plants has to supply power needs of the station (station load) such as hot water pumps, draft air pumps, coal pulverisers etc. to reduce these losses, hot water (steam) is supplied to

industries where steam plays a major role in their production process such as paper mills, chemical plants, food canning etc. The schematic diagram of a steam or Coal fired thermal power plant is given in Figure 3.

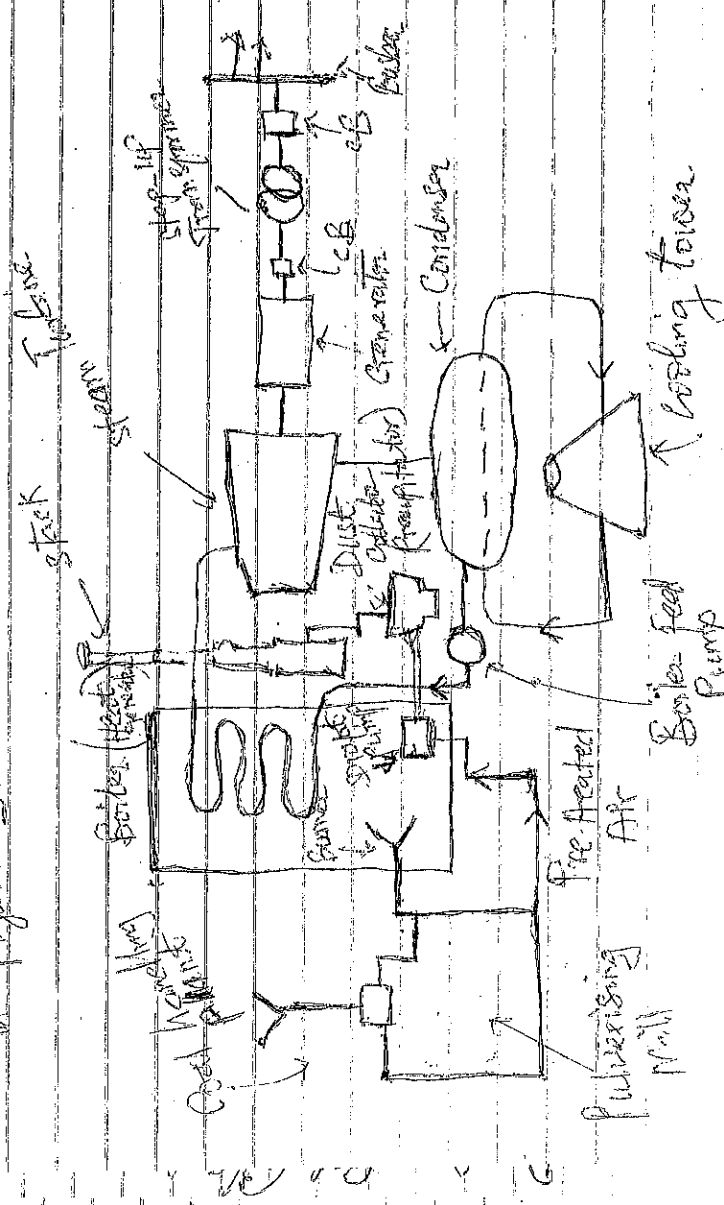


Figure 3: Schematic diagram of a coal fired thermal power plant.

* Super heating machines
 ↑ Gm. Sp. H.P. 13
 attached in p. 13

* Large Unit consume less fuel / kWh.
 * Lower labour cost
 * Low maintenance expenditure
 * Cost saving up to 15% for 100 - 350 MW plants
 * Plants can be started.
 * Up to 1000 MW units have been achieved

The ~~thermal~~ (steam) power plant operates on the Rankine Cycle
 boiler - turbine - condenser - pump where by Mechanical work is extracted from a fluid as it moves between a heat source and a heat sink. The Rankine Cycle is ~~the~~ used in both Coal and Nuclear power generation stations. The Rankine Cycle is modified to include air preheaters and steam super heaters to obtain steam at the highest possible temperature and pressure. To take advantage of economy of scale (Cost reduction through increase in production) large size plants are preferable, thus practice has led to lowering the cost per kilowatt of power generation.
 BASIC COMPONENTS OF A COAL FIRED THERMAL POWER PLANTS.

1. Boiler :- This is the steam generator, where feed water and the water supplied from external source is converted into wet steam. There are two types of boilers, the water tube boiler and fire tube boiler. The former has a circulating hot water in side tubes while the latter has circulating hot gases inside tube.

2. Steam Turbine :- In the steam turbine work is done by expansion of steam. The expansion occurs when the wet gases exert a specific temperature and pressure (85 kg/cm² at 600°C). The turbine converts superheated steam into Mechanical energy. There are two types of steam turbines, the impulse type and reaction type.

3. The Condenser :- The condenser is the equivalent where the exhaust steam (gas) is converted into feed water or given off as rejected in the cooling towers or even discharge into the stream.

4. Stack and Precipitator :- Act as electrostatic precipitator by removing dust particles from the hot exhaust gases inside the condensation chamber.

5. Superheaters in a modified Rankine Cycle the

Superheater Unit is in addition to further raise the temperature of steam by absorbing heat from exhaust (flue) gases from the combustion chamber/boiler.

6. Coal Handling Plant: In a coal handling plant, coal from the storage site is exposed to sun to remove wet (increase its energy content (KJ) i.e. to raise the calorific value. The dried coal is then fed into the boiler furnace.

Advantages & Disadvantages of Thermal Power plants?

(a) Advantages:

1. Low initial/capital cost compared to nuclear and hydro power plants.
2. Requires less land area compared to hydro power plant.
3. Thermal power plants rapidly to change in load.
4. Portion of heat generated can be used to process heat in some industries.

(b) Thermal plants require less gestation period.

6. Fuel used (Coal) is cheaper.

(c) Disadvantages:-

1. Thermal plant requires large quantity of water for boiler and condenser cooling.

2. Fuel transportation cost is high especially when plants are situated far from coal mines.

3. Thermal power plant has low overall efficiency.

4. Thermal power plant pollutes the atmosphere by fumes, dust and residues of pulverised coal.

5. Problem of waste disposal (ash) 2-3 tonnes is generated in a large power plant.

NUMERICAL EXAMPLE:-

A thermal power plant spends \$848 million/year as cost of coal. If the cost of coal is 40,000/tonne and its calorific value is 5000 kcal/kg. The efficiency of the station is 30%. Calculate:

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- a) Quantity of Coal Consumed Per Annum
- b) The Heat input for the Combustion per year
- c) The heat output per year
- d) Electrical energy generated (Kwh) per year
- e) Average generated (MW) by the station per year.

Given: Total cost of Fuel (coal) per year = Rs 848,000,000
Combined station efficiency = 30% = 0.3 pu
Energy content per Kg of Coal (Calorific Value)

$$= 5000 \text{ Kcal/Kg.}$$

$$\text{Unit cost of Fuel (coal)} = \text{Rs } 40,000 \text{ per Tonne.}$$

- c) Quantity of coal consumed per annum = Total cost

Unit cost / Tonne

$$= \frac{\text{Rs } 848,000,000}{40,000}$$

$$= 21,200$$

- b) Heat input = Calorific Value $\times$ quantity consumed in annum
in a year

$$= 5000 \times 21,200 \times 1000 \text{ (Kcal/year)}$$

$$= 106,000,000,000 \text{ Kcal/year}$$

$$= 106 \times 10^9 \text{ Kcal/year,}$$

- d) Heat output per year = Heat input $\times$ Efficiency of plant

$$= 106 \times 10^9 \times 0.3$$

$$= 31.8 \times 10^9$$

$$= 3.18 \times 10^{10} \text{ Kcal/year}$$

- e) Electrical energy generated (Kwh)

$$\text{Energy (Kwh)} = 860 \text{ Kcal (Energy Kcal)}$$

$$\therefore \text{Electrical Energy (Kwh)} = \frac{31.8 \times 10^{10}}{860} = 36.98 \text{ Mwh}$$

$$= 36.98 \text{ Mwh}$$

- a)

Average Electrical Power generated by the plant

$$= \frac{\text{Electrical energy (Kwh) generated}}{\text{Hours of operation}}$$

$$= \frac{36.98 \text{ Mwh}}{24 \text{ hours}}$$

$$= 1.54 \text{ MW}$$

(16)

THE NUCLEAR POWER STATION

The nuclear power plant operation is similar to the normal steam/thermal power plant. The difference is only in the fuel used to generate heat in order to boil water.

In nuclear plant the sustained fission of nuclear material is used to generate heat. In nuclear

plants, the nuclear reactor act as a furnace where heat is generated by splitting or fissioning the nucleus

of fissionable materials particularly Uranium 235 (Sp235). When the nucleus of Uranium (U_{235}) absorbs a neutron

at high velocity the heavy Uranium nucleus splits into two or more lighter nuclei releasing enormous

amount of heat energy. The reaction also releases

and harmful radiations (gamma rays) and some free

neutrons. A portion of the released neutrons are then

absorbed by the available fissionable nuclei and trigger

further fission which in turn releases more neutrons.

This ensures a sustained nuclear reaction known

as the chain reaction. Natural Uranium ore

contains very small amount of the fissionable U_{235} atoms

(about 0.7%) the Uranium has to be refined and enriched

to attain the desired level of concentration of the U_{235}

atoms suitable for any desired application (Commercial power

plants fuel, nuclear weapons). The concentration of

or U_{235} atoms by removing U_{238} atoms so that it can

be used as a nuclear fuel in nuclear power plants.

Nuclear research centres or nuclear weapons are

called nuclear enrichment. The enriched

Uranium powder is then formed into metal bars and

are bundled into fuel assemblies to be used into

the reactor as the nuclear fuel. Figure 4.1 shows

the layout diagram of a nuclear power plant.

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Exhibits

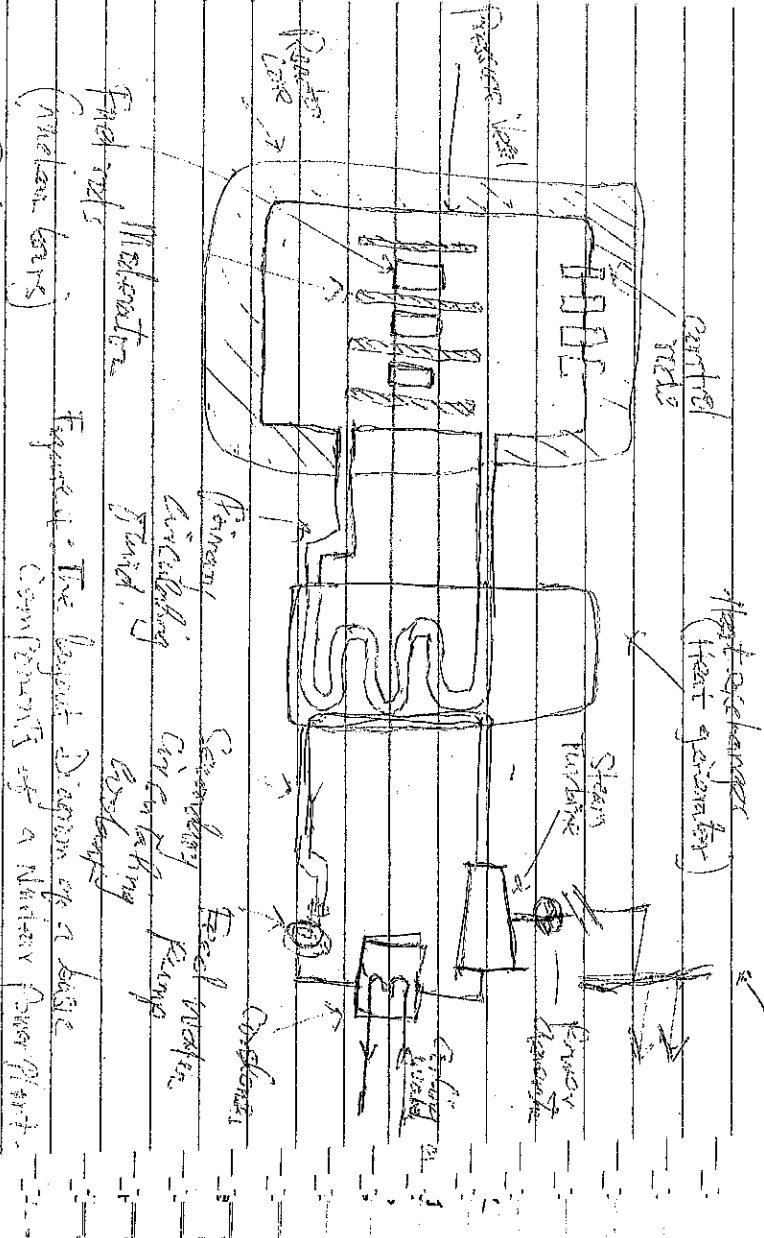


Figure 1: The layout diagram of a basic configuration of a Nuclear Power Plant.

COMPONENTS OF THE NUCLEAR POWER PLANT

- (1) **Reactor Core**:- This is the main component of a nuclear power plant. It is the apparatus in the nuclear reaction takes place the nuclear fuel is subjected to controlled nuclear fission to release high speed fission products. The reactor consists of a cylindrical pressure vessel which houses the uranium bars, moderators and control rods.
- (2) **Moderators**:- The moderators are used to slow down the fast high speed neutrons so as to increase the likelihood of fission. Examples of moderator materials are: heavy water and graphite or beryllium.
- (3) **Control Rods**:- These are devices used within the nuclear reactor chamber to absorb excess neutrons to control the rate of a nuclear reaction chain reaction. They are rods inserted into the reactor core and are made of materials such as 'Boron steel', cadmium, hafnium. They are strong neutron absorbers and are thus used to regulate and control the rate of reaction.

(13)

Thus raising or lowering the control rods regulate the power output of a nuclear power plant.

(4) the coolant:-

The heat generated in the reactor is removed by the coolant. The primary coolant absorbs heat generated in the reactor and transfers it to the steam generator.

Bronghts are liquid sodium, sodium alloy helium. The secondary coolant transfers the pressurised steam to be fed through the turbine and drives the electrical generator. The water-steam mixture is then cooled down in a condenser.

(5) Heat Exchanger:- The primary coolant gives up the heat in the heat exchanger/heat generator where steam is generated.

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LOAD CHARACTERISTICS AND ECONOMICS OF POWER GENERATION

In power generation economics, the system design must be done in compliance with strict economic considerations. The power is designed and operated on the basis of the most economical condition as profitability is one of the most important factors in consideration. The objective is to bring the cost of electrical energy produced to a minimum value.

LOAD CURVES OF A GENERATING STATION

The total power requirements of a given generating station can be estimated using a number of curves and diagrams which includes — the load curve and the load duration curve.

The load curve — This is a two-dimensional graph showing the variation of the station load with respect to time. The load curve also called the load profile diagram drawn between the load (MW) against time in sequential order. Thus the area under a load curve always represents the total energy consumed by the load in the specified period of time (daily, weekly or annually.)

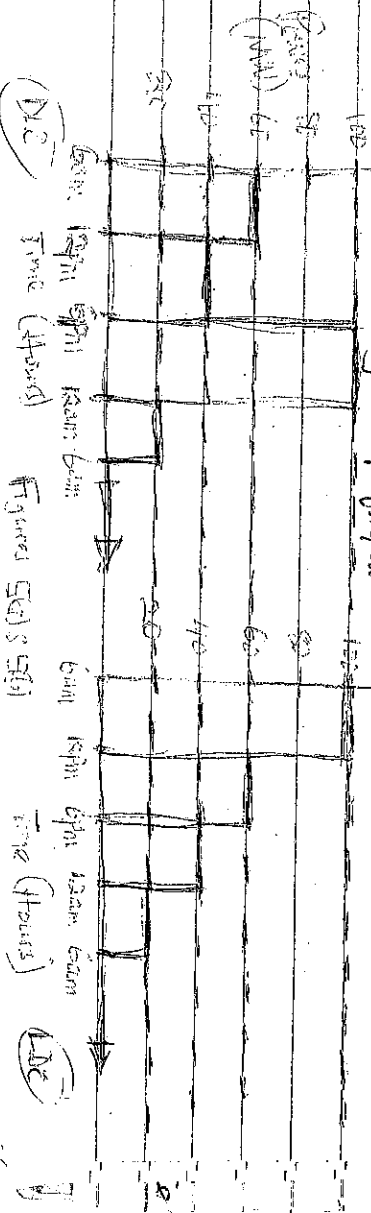
The load duration curve (LDC) — The load duration

curve is similar to the load profile. It also represents the instantaneous load against time but the ordinates are rearranged in order of magnitude and not time sequence.

i.e. the vertical coordinates that represents the power consumed is arranged in the system in order of magnitude

with the greatest to the smallest plotted value plotted before the smallest from left to right. Figure 5(a) and 5(b) shows the Daily load curve (DLC) load profile and load duration curve. (LDC). For a daily load curve the total area under the curves —

Figure 5(a) & 5(b)
Daily load profile/curve Load duration curve



(20)

represents the total units of energy consumed in a day (kWh).

IMPORTANCE OF LOAD CURVES

The load curves (load profile and load duration curve) have significant importance in power generation system, they provide the following information readily.

- (i) It indicates the peak load on a power station and the time it is occurring by inspection. It also shows the variation of load on the station during different hours of the day, days of week or month, thereby the area under the load curve gives the total number of units of electricity generation in any given period of time. i.e. (Units of Energy generated) = Area under the curve (kWh) = kilowatts × time (hrs).

- (ii) Area under the load curve divided by the total time of operation gives the average load on the station (MW).

- (iii) The highest point on the load curves represents the maximum demand on the station or the average

- (iv) The load curves provide information for the power selection of the sequence at which generating units will be operated and the selection of the economic sizes of generating units.

(v) It can be used in estimating the generating cost and preparing the operation schedule of the power station.

- (vi) The ratio of the area under the load curve to the total area of the rectangle in which it is contained gives the load factor. Thus Load Factor (LF) = $\frac{\text{Area under the curve (kWh)}}{\text{Total area of the rectangle in which the load curve is contained}}$

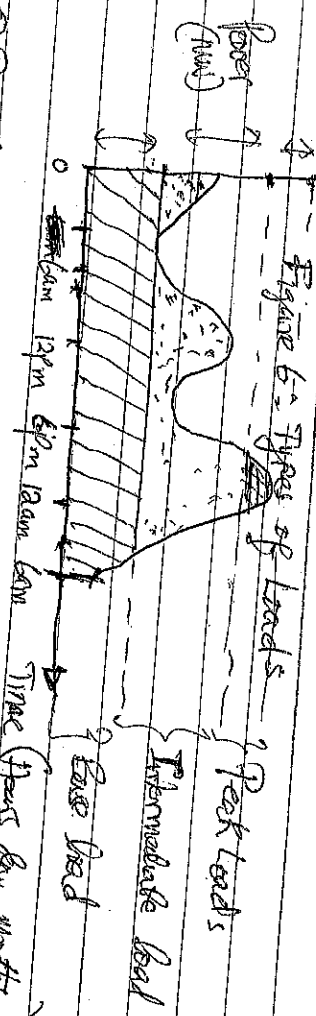
$$\therefore LF = \frac{\text{Average load}}{\text{Maximum Demand}}$$

$$= \frac{\text{Area (kWh) under the load curve}}{\text{Maximum Demand (MW)}}$$

$$= \frac{\text{Average load} \times 24 \text{ hrs}}{\text{Maximum Demand} \times 24 \text{ hrs}} = \frac{\text{Average Load}}{\text{Maximum Demand}}$$

(21) TYPES OF ELECTRICAL LOADS

The amount of electrical energy or electrical power consumed by a particular device or a whole community is referred to the electrical load or energy demand. The loads intermediate loads are classified three basic types: Base in Figure 6.



① Base loads:- This is the load which the demand never falls is supplied 100% of the time. The type Base loads are normally

supplied by nuclear or lighter power plants. Base loads are constant loads that does not vary with time the occur on the station through the period of operation

② Intermediate loads:- This loads represents the region between the base and the peak load as shown in the curve.

③ Peak load:- This is the power demand above the intermediate loads on the station. The Peak load forms a small portion of the total load on the station.

Factors AFFECTING COST of Power GENERATION cost

The following factors are used in determining the cost of electrical energy per kilowatt-hour (kWh): Average load, Maximum Demand, Diversity Factor, Plant capacity Factor,

① Average load:- This is the average load station in a given period (day, month or year),

(i) Daily average load = No. of units (kWh) generated in a day

(ii) Monthly average load = No. of units (kWh) generated in a month

(iii) Yearly average load = No. of units (kWh) generated in a year

(20)

② Maximum Demand (MD): - This is the greatest demand or load on a power station during a given period. The knowledge of the max. demand helps in determining the installed capacity of the plant.

③ Demand factor: This is the ratio of the maximum demand on the station to its connected load. The value of the demand factor is less than unity because the maximum demand on a station is less than its connected load or total load.

$$\text{Demand Factor} = \frac{\text{Max. Demand}}{\text{Connected load}}$$

④ Load Factor (LF): - This is the ratio of the actual energy generated to the energy which could have been generated if the plant is operated continuously under maximum demand condition. It is the ratio of average load to the maximum demand.

$$\text{Daily Load Factor (LF)} = \frac{\text{Average load}}{\text{Max. load}} = \frac{\text{MD Consumed} \times 24 \text{ hrs}}{\text{MD} \times 24 \text{ hrs}}$$

The value of Load Factor is usually less than unity, however higher load value of the load is better for power companies.

THE IMPORTANCE OF LOAD FACTOR IN POWER PLANT

OPERATION

The load factor is an index of the proportion of the whole time a generating plant is being used in its full capacity i.e. plant operated under maximum load. Though generating plants are selected base on the peak load that will occur on the station, but it seldom happens that the generator is load to its maximum capacity throughout. Whether a generator unit is run at full load or not, fixed charges (interest, depreciation, taxes, insurances) always apply. This means the generating sets are costing more to the owner whether working or idle. Therefore from economic point of view, the equipment earns profit only when fully loaded. Hence it is desirable to obtain high load factor. If the Load factor is poor, it means less number of units are generated, then charges per kWh would be higher.

Examples

- ① The maximum demand on a power station is 150 MW. If the annual load factor is 40%. Calculate the total energy generated in the year.

$$MD = 150 \text{ MW}, LF = 40\% = 0.4$$

Hours of operation in a year = $365 \times 24 = 8760 \text{ hrs.}$

$$\text{Total Energy generated} = \text{Average load} \times \text{period of operation}$$

$$= \text{Average load} \times 8760 \text{ (hrs)}$$

$$\text{But from load factor (LF) = Average load}$$

$$\text{Max. Demand}$$

$$\Rightarrow \text{Average load} = \text{Load Factor} \times$$

$$\text{Max. Demand}$$

$$\therefore \text{Energy generated} = \text{Load Factor} \times \text{Max. Demand}$$

$$\text{on the station} \times 8760 \text{ hrs}$$

$$= 150 \text{ MW} \times 0.4 \times 8760$$

$$= \underline{\underline{520,400 \text{ MWh}}}$$

- ② Estimate the maximum demand for a load which consumes 500 kWh of electricity per day at a load factor of 40%. If the consumer increases the maximum load factor to 60% without increasing the maximum demand, determine the amount of electrical energy used.

$$\text{③ Load factor (LF) = Energy Consumed}$$

$$\text{(Daily)}$$

$$\text{Maximum Demand} \times 24 \text{ hrs}$$

$$= 5000 \text{ kWh}$$

$$40\% = 0.4 = \frac{5000}{\text{Max. Demand} \times 24 \text{ hrs}}$$

$$\Rightarrow MD = \frac{5000}{0.4 \times 24}$$

$$MD \times 24 = 520.8 \text{ kW}$$

$$\text{④ New load factor } 60\% = 0.6 = \underline{\underline{520.8 \text{ kW}}}$$

$$\text{New Energy Consumed} = \text{load factor} \times \text{Max. Demand}$$

$$\times 24 \text{ hrs,}$$

$$= 0.6 \times 520.8 \text{ kW} \times 24 \text{ hrs}$$

$$= \underline{\underline{7500 \text{ kWh}}}$$

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- ③ A generating station has a connected load of ~~120 MW~~ ^{120 MW} and supplies a maximum load of 60 MW. The number of units generated in a year is 48×10^7 kWh. Calculate the load factor and the demand factors of the generating station.

(a) Load Factor (LF) = $\frac{\text{Total energy generated in the year}}{\text{Maximum Demand} \times \text{Hours in a year.}}$

$$= \frac{48 \times 10^7}{60,000 \times 8760}$$

$$= 0.913$$

(b) Demand Factor = $\frac{\text{Maximum Demand}}{\text{Connected load}}$

$$= \frac{60,000}{120,000}$$

$$= 0.5$$

- ④ A power plant has the following daily load cycle as follows:

Load (MW)	20	40	60	20	50	20
Time (hrs)	6-8	8-12	12-16	16-20	20-24	24-6

Plot the load curve and the load duration curve. Calculate the energy generated per day, average load and the station load factor.

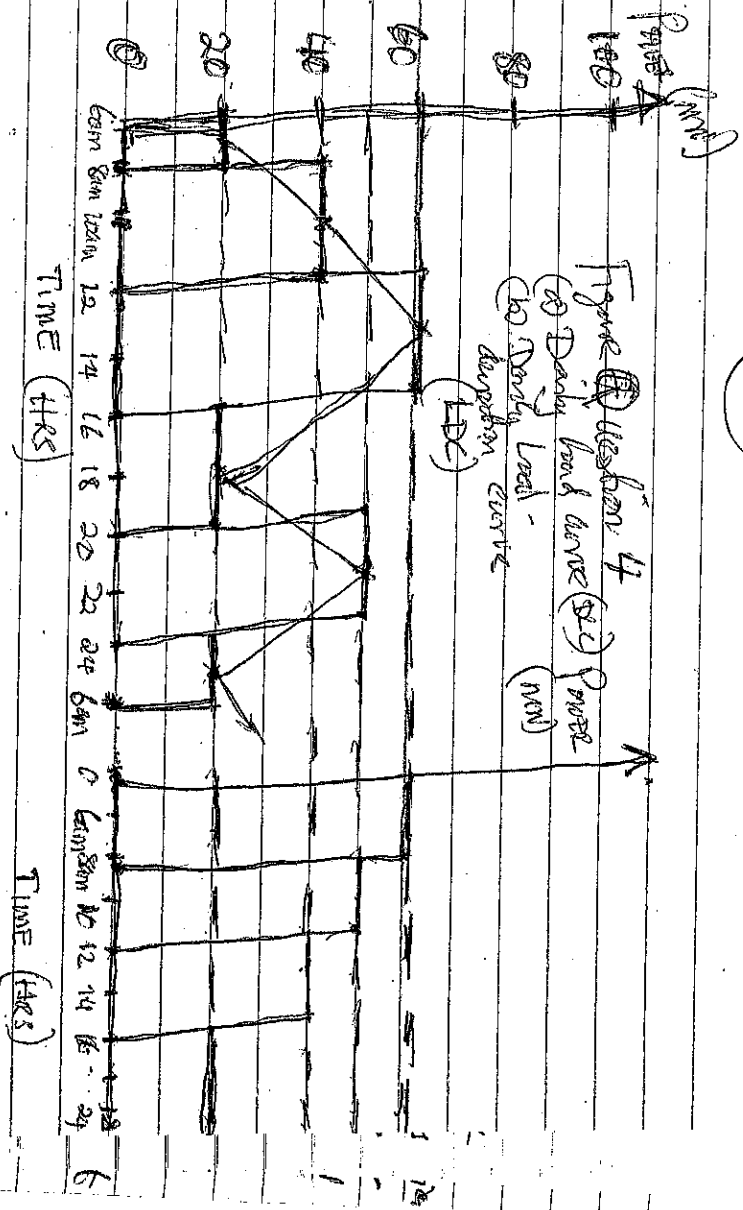
Energy Generated per day = $\frac{\text{Area Under the daily load curve}}{\text{Load curve}}$

= $\frac{\text{Area Under the load duration curve.}}$

Daily Energy Generated

by power plant = $(20 \times 12) + (40 \times 4) + (60 \times 4) + (20 \times 4) + (50 \times 4) + (60 \times 4)$
 $= 240 + 160 + 240 + 240 + 240 + 240$
 $= 840 \text{ MWh}$

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Exercise ~~Assignment~~

1. A Power Supply has to meet the following load demand:-

Alkuvu	300km between 6am and 8pm and 150km between 8pm and 6am
Garden City	100km between 9am and 10pm
Mure Pabys	100km between 6am and 3pm
Fed Polytechnic	150km between 6am and 9am and 100km between 9am and 8pm

- (a) Calculate Plot the daily lead curve and the load duration curve
- (b) Calculate the load Factor off the spinin

(26)

THE PER UNIT SYSTEM ANALYSIS

The Per Unit Value is a Unitless Value. It is a ratio between values of two quantities of the same unit. The use of Per Unit (p.u.) notation helps to simplify calculations in power system especially in interconnected systems where several network areas or sections are connected together through transformers. The per unit value of a quantity is defined as the ratio of the actual quantity to an arbitrarily chosen value of that quantity in the same unit. Thus,

$$\text{Per Unit Value} = \frac{\text{Actual Value in the same unit}}{\text{Base Value in the same unit}}$$

Before the introduction of the per unit method of analysis, power system values are calculated using the absolute values of the (Voltage, Current, Impedance and Power). Consider the power network diagrams shown in Figures 6(a) and 6(b), the number of equations required to analyse each network would depend on the number of operating voltage levels involved in each diagram.

Busbar

Generating station

Figure 6(a): Power Network with single operating Voltage.

Fig 6(a) Single voltage level (No transformation is involved), it will require only one equation to calculate and determine the electrical quantities in the network.

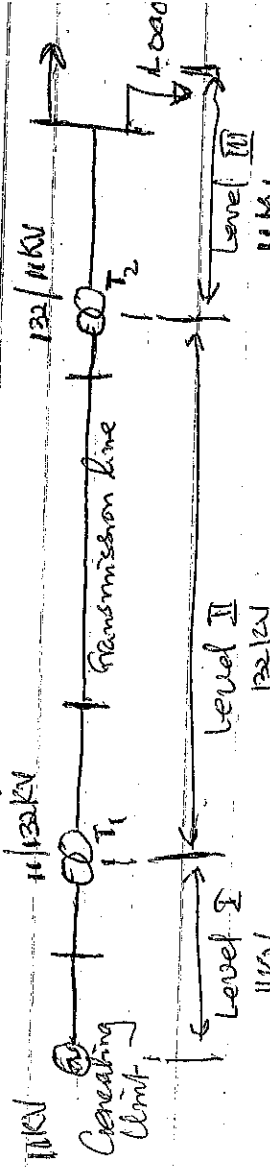


Figure 6(b): Power network with different operating Voltage levels.

1.9. With 2.5%
 Express absolute value in
 as per unit of base value
 as reference value
 as reference base.

Since ~~there~~ there are three voltage levels in the network it will require three equations in order to analyse and determine the network parameters. Thus in a large power network (modern power network) the network operates with several voltage levels this results into several equations required to analyse the performance of the network. Several equations requires more computation time and computer memory. Due to this limitation, the absolute system of power analysis is being replaced with the per unit system of analysis.

RELATIONSHIP BETWEEN THE ABSOLUTE VALUES AND THE PER UNIT VALUES.

The per unit value of any electrical quantity can be obtained as a ratio of the absolute value divided by the base value or reference value as follows:

$$\text{Per Unit Value} = \frac{\text{Absolute Value}}{\text{Reference or Base Value}}$$

Where the absolute value is the actual value of the quantity and the base value is selected arbitrarily as a positive whole number of the same unit with the actual quantity.

Consider two electrical machines in a network.

Machine 1: V_1 (Volts) and I_1 (Amperes)

Machine 2: V_2 (Volts) and I_2 (Amperes)

Let choose a base value for voltage and current for the network as follows:

The $V_b \Rightarrow V_b$ and $I_b \Rightarrow I_b$
~~By these~~ per unit values for the two machines are:

(28)

① Per Unit (P.u) Voltage of machine 1;

$$V_1(\text{P.u}) = \frac{V_1}{V_b} = \frac{V_1}{V_1} = 1.0 \text{ P.u}$$

$$I_1(\text{P.u}) = \frac{I_1}{I_b} = \frac{I_1}{I_1} = 1.0 \text{ P.u}$$

② For Machine 2:

$$V_2 \text{ per unit} = \frac{V_2}{V_b} = \frac{V_2}{V_1} (\text{P.u})$$

$$I_2 \text{ per unit} = \frac{I_2}{I_b} = \frac{I_2}{I_1} (\text{P.u})$$

THE PER UNIT REACTANCE

The reactance of large or high power electrical machines (motors, generators and transformer) is always very small compared to the reactance of the machines it is therefore neglected in carrying out analysis.
By definition:

Reactance (X) of an electrical equipment is given by:

$$X(\text{ohms}) = \frac{V(\text{Volts})}{I(\text{amps})} = \frac{V}{I}$$

For the two machines their base and per unit reactance values can be obtained as follows:-

$$\left(\frac{\text{Absolute Value}}{\text{Value}} \right) X_b(\Omega) = \frac{V(\text{Volts})}{I(\text{amps})} = \frac{V_1}{I_1} = \frac{V_b}{I_b} = X_b(\Omega)$$

= The base value of reactance for machine 1 = $X_b(\Omega)$

$$\text{For Machine 2} \Rightarrow X_{2, \text{pu}} = \frac{V_2}{I_2} = \frac{V_b}{I_b} (\Omega)$$

∴ Per unit reactance of the machines:

(29)

$$\textcircled{1} X_1 (P.U) = X_1 (\text{actual}) = \frac{X_1 (\Omega)}{X_{base}} = \frac{X_1 (\Omega)}{X_{base} \times X_{base}^{-1} \times 10^3}$$

$$\textcircled{2} X_2 P.U = \frac{X_2}{X_b} = \frac{X_2}{X_1' (P.U)}$$

Generally: The Per Unit Values for the current voltage and reactance of electrical equipment is given by:

$$\left\{ \begin{aligned} I_{P.U} &= \frac{I_{(\Omega)}}{I_{base}} \Rightarrow \frac{I_{(\Omega)}}{I_{base}} \\ V_{P.U} &= \frac{V(\text{Volts})}{V_b(\text{Volts})} = \frac{V}{V_b} \quad \text{--- (1)} \\ X_{P.U} &= \frac{X_{(\Omega)}}{X_b} \end{aligned} \right\}$$

From Equation 1 above the P.U reactance of electrical Machines can also be expressed in terms of base current and base voltage, thus:

$$X_{P.U} = \frac{X_{\Omega}}{X_b} \quad \text{But } X_b = \frac{V_b}{I_b}$$

$$= \frac{X_{\Omega}}{X_b} = X_{\Omega} \left(\frac{I_b}{V_b} \right)$$

$$X_{P.U} = X_{\Omega} \left(\frac{I_b}{V_b} \right) P.U \quad \text{--- (2)}$$

Using Equation 2, the per unit reactance of all power equipment can be determined.

TRANSFORMERS

A transformer is always rated in (Volts Amperes) i.e. VA for small transformers and KVA or MVA for large power transformers. We can express the power rating for a transformer in per unit as follows:

$$\begin{aligned} \text{Per unit (KVA/MVA)} &= \frac{\text{Actual KVA}}{\text{Base KVA}} \\ &= \frac{\text{KVA (actual)}}{\text{KVA (base)}} \end{aligned}$$

The per unit reactance can be also be expressed in terms of base KVA or MVA and base KV of the network in which the transformer is connected. From Equation 2 above:

$$\begin{aligned} X_{pu} &= X_{(2)} \left[\frac{I_b}{V_b} \right] = X_{(2)} \left[\frac{I_b V_b}{V_b V_b} \right] = X_{(2)} \left[\frac{I_b V_b}{V_b^2} \right] \\ &= X_{(2)} \left[\frac{I_b V_b}{V_b^2} \right] \end{aligned}$$

Divide Numerator and Denominator by 10^6

$$X_{pu} = \left[\frac{V_b I_b / 10^6}{V_b^2 / 10^6} \right] X_{(2)} = \left(\frac{\text{MVA}_b}{KV_b^2} \right) X_{(2)}$$

$$X_{(pu)} = \left(\frac{S_B}{V_B^2} \right) X_{(2)} \quad \text{--- (3)}$$

(31)

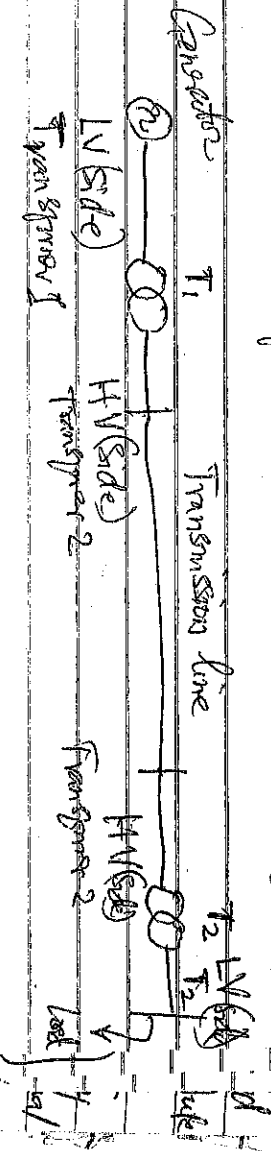
RULES FOR THE SELECTION OF BASE VALUES WHEN A TRANSFORMER IS CONNECTED IN A NETWORK.

① When a transformer is connected in a network, two set of base values (S_B and V_B) are selected one set for the low voltage (LV) side and another set for the high voltage (HV) side.

② Common base power is selected for the entire network.

③ Two different base voltages must be selected for LV and HV side of the transformer in such a way so that their ratio must be equal to the transformation of the transformer.

④ If two transformers are connected in series in a network, the number of base voltages selected are three with one common to the two transformers as shown in Figure 7.



∴ The three bases to be selected are:

$$LV_1, HV_1(T_1), LV_2$$

BASE VALUES FOR THREE-PHASE POWER SYSTEM

The base values considered when the per unit calculations is carried out are the three-phase (3- ϕ) power and the line voltages (V_L) respectively.

(32)

For a Single-phase System;

$$X_{b'}(1-\phi) = KV_b^2(1-\phi) = \frac{V_b^2(1-\phi)}{\frac{MVA_b(1-\phi)}{S_b^2(1-\phi)}}$$

For a three-phase System

$$X_b(3-\phi) = \frac{KV_b^2(\text{line})}{MVA_b(3-\phi)}$$

$$\text{But } V_b = \sqrt{3} V(1-\phi)$$

$$\therefore X_b(3-\phi) = \frac{KV_b^2 \text{ line}}{MVA_b(3-\phi)} = \frac{(\sqrt{3} KV_b(1-\phi))^2}{3 \times MVA_b(1-\phi)}$$

$$= \frac{3 \times KV_b^2(1-\phi)}{3 \times MVA_b(1-\phi)} = \frac{KV_b^2(1-\phi)}{MVA_b(1-\phi)}$$

Hence; ~~$X_b(3-\phi)$~~

$$\boxed{X_{b'}(3-\phi) = X_b(1-\phi)} \quad (33)$$

CHANGE OF BASE VALUES

When a new equipment are added to a power net or grid the per unit reactance of the network ~~this~~ changes due to change the base values. To express the ~~(to obtain)~~ the new P.u. reactance values in terms of the old P.u. reactance values, the following procedure is used:-

(33)

From Equation 3, the dd per unit reactance ~~can~~ is given by:

$$X_{pu}^{(old)} = X_{(2)} \frac{MVA_b^{(old)}}{KV_b^2 (old)} = \left[\frac{MVA_b^{(old)}}{KV_b^2 (old)} \right] X_{(2)}$$

$$= \left[\frac{S_b^{(old)}}{V_b^2 (old)} \right] X_{(2)}$$

$$\text{Similarly, } X_{pu}^{(new)} = X_{(2)} \left[\frac{MVA_b^{(new)}}{KV_b^2 (new)} \right] = \left[\frac{S_b^{(new)}}{V_b^2 (new)} \right] X_{(2)}$$

$$\text{New, } \frac{X_{pu}^{(new)}}{X_{pu}^{(old)}} = \frac{X_{(2)} \left(\frac{MVA_b^{(new)}}{KV_b^2 (new)} \right)}{X_{(2)} \left(\frac{MVA_b^{(old)}}{KV_b^2 (old)} \right)} = \frac{X_{(2)} \left(\frac{S_b^{(new)}}{V_b^2 (new)} \right)}{X_{(2)} \left(\frac{S_b^{(old)}}{V_b^2 (old)} \right)}$$

$$\frac{X_{pu}^{(new)}}{X_{pu}^{(old)}} = \frac{S_b^{(new)}}{S_b^{(old)}} \frac{V_b^2 (old)}{V_b^2 (new)}$$

$$\frac{X_{pu}^{(old)}}{X_{pu}^{(new)}} = \frac{S_b^{(old)}}{S_b^{(new)}} \frac{V_b^2 (new)}{V_b^2 (old)}$$

$$\frac{X_{pu}^{(new)}}{X_{pu}^{(old)}} = \frac{S_b^{(new)}}{S_b^{(old)}} \left[\frac{V_b^{(old)}}{V_b^{(new)}} \right]^2$$

$$\frac{X_{pu}^{(new)}}{X_{pu}^{(old)}} = \frac{S_b^{(old)}}{S_b^{(new)}} \left[\frac{V_b^{(old)}}{V_b^{(new)}} \right]^2$$

(5)

Hint: $M \rightarrow N(\text{new})$ Numerator

Generally whenever the base values are changed, it is the p.u. values that changes and not the absolute or actual values of the equipment.

Example 1: An equipment is having a per unit value of 10 p.u. reactance on a 100 MVA, 10 kV base values. Find the new per unit reactance value of the equipment on a new base values of 10 MVA and 100 kV.

$$X_{p.u}^{\text{new}} = X_{p.u}^{\text{old}} \left[\frac{MVA_{\text{old}}}{MVA_{\text{new}}} \right] \left[\frac{KV_{\text{old}}}{KV_{\text{new}}} \right]^2$$

$$= 10 \left(\frac{10}{100} \right) \left(\frac{10}{100} \right)^2 = 10 \left(\frac{10}{100} \right) (0.01)^2$$

$$= 10 (0.01) (0.01)$$

$$= 0.01 \text{ p.u.}$$

Example 2: A generating station has a base values of 30 MVA and 11 kV with a per unit reactance of 1.2 p.u. - What is the new p.u. reactance if the base of the generating station is selected as 75 MVA, 13.2 kV.

$$X_{p.u}^{\text{new}} = X_{p.u}^{\text{old}} \left[\frac{MVA_{\text{old}}}{MVA_{\text{new}}} \right] \left[\frac{KV_{\text{old}}}{KV_{\text{new}}} \right]^2$$

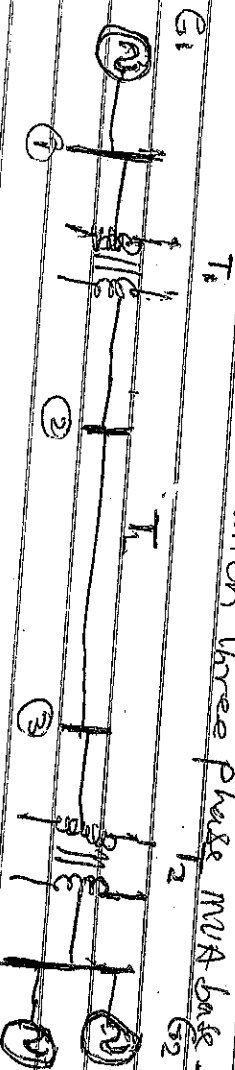
$$= 1.2 \left[\frac{30}{75} \right] \left[\frac{11}{13.2} \right]^2 = \frac{25}{12} = 2.08 \text{ p.u.}$$

(34)

Example 3: Find the per unit reactance of an alternator with base values of 30 MVA, 11 KV with absolute reactance value of 8Ω .

$$X_{pu} = X_{\text{abs}} \frac{\text{MVA}_b}{\text{KV}^2} = \frac{X_{\text{abs}}}{X_b} = \left(\frac{30}{11^2} \right) = 1.49 \text{ p.u.}$$

Example 4: obtain the per unit equivalent reactance diagram for the single-line diagram of a power system network as shown in Figure 8. choose 30 MVA as a common three phase MVA base.



$G_1: 30 \text{ MVA}, 10.5 \text{ KV}, X'' = 1.6 \Omega/\text{phases}$

$G_2: 15 \text{ MVA}, 6.6 \text{ KV}, X'' = 1.2 \Omega/\text{phases}$

$G_3: 25 \text{ MVA}, 6.6 \text{ KV}, X'' = 0.56 \Omega/\text{phases}$

$T_1 = 15 \text{ MVA}, 11/33 \text{ KV}, X = 15.2 \Omega/\text{phase}$

$T_2 = 15 \text{ MVA}, 33/6.6 \text{ KV}, X = 15.2 \Omega/\text{phase on HV side}$

Transmission line 1: $X_L = 20.5 \Omega/\text{phase}$

Solution: Assumptions

- (i) Line capacitances and resistance are neglected so that the impedances of the various components are represented by their reactances only.
- (ii) The reactance diagram is meant for short circuit analysis so that current drawn by static loads are small and ignored. Hence loads are ignored.

35

Convert

To all the reactances from the given absolute (ohmic) values to per unit form.

① Choose a common base for MVA and

Voltage for the network.

② Base power = 30 MVA = MVA_B

Base Voltage = Three operating voltage levels. So three different base voltages required as follows:

(a) 33 kV = on the transmission line i.e. i.e. HV side (T_1 and T_2) = kV_{B1}

(b) 11 kV = on the generator 1 circuit i.e. LV side (T_1) = kV_{B2}

(c) 6.2 kV = on the circuit of generator 2 and 3 i.e. LV side (T_2)

③ Computation of the per unit reactances of the components:

(a) Transmission line $X_T(\%) = 20.5\%$

from Equation (3) $X_T(p.u.) = \frac{MVA_B}{KV_B^2} \times \%$

$$X_T(p.u.) = \frac{30}{(33)^2} \times 20.5\%$$

$$= 0.564$$

(b) Transmission (T_1) & $X_{T_1} = 15.2\%$ (HV side)

$$X'_{T_1}(p.u.) = \frac{30}{(33)^2} \times 15.2\%$$

$$= 0.418$$

(36)

Transformer 2 (T_2): $X_{T_2} = 16 \angle 0^\circ$

$$X_{T_2} (P.u) = \frac{30}{33^2} \times 16$$

$$= 0.44$$

Generator 1 (G_1): $X_{G_1} = 1.6 \angle 0^\circ$

$$X_{G_1} (P.u) = \frac{30}{(11)^2} \times 1.6$$

$$= 0.396$$

Generator 2 (G_2): $X_{G_2} = 1.2 \angle 0^\circ$

$$X_{G_2} (P.u) = \frac{30}{(6.2)^2} \times 1.2$$

$$= 0.936$$

Generator 3 (G_3): $X_{G_3} = 0.56 \angle 0^\circ$

$$X_{G_3} (P.u) = \frac{30}{(6.2)^2} \times 0.56$$

$$= 0.437$$

$$X_{T_1} (P.u) = 10.5 = 0.96$$

$$X_{T_2} (P.u) = \frac{6.6}{6.2} = 1.06$$

$$X_{G_3} (P.u) = \frac{6.6}{6.2} = 1.06$$

37

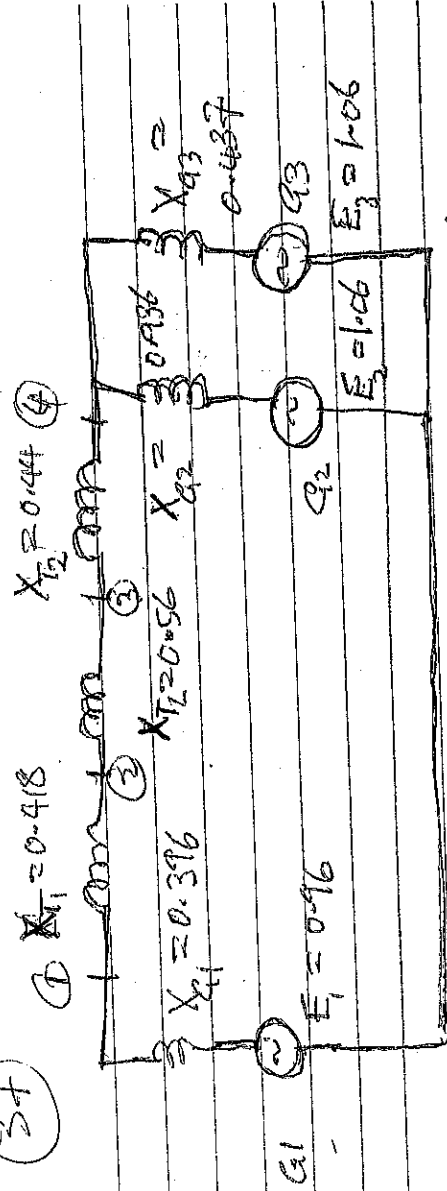


Figure Q4: Reactance Diagram of the Network for Q4.

PERFORMANCE OF TRANSMISSION LINES

The performance of overhead transmission line can be obtained by the evaluation of two main electrical parameters of the transmission line:

- ① Efficiency of the line
 - ② Voltage regulation of the line
- The efficiency of the transmission line evaluated in terms of the real or active transmitted and received as follows:-

$$\text{Percentage efficiency} = \frac{\text{Power output}}{\text{Power input}} \times 100\%$$

(N)

Since, $\text{Power output} = \text{Power output} + \text{Power losses on the line}$

$$\% \eta = \frac{\text{Power output}}{\text{Power output} + \text{losses}} \times 100\%$$

$$= \frac{\text{Power out}}{\text{Power out} + \text{losses}} \times 100\%$$

Similarly $\text{Power output} = \text{Power input} - \text{losses}$

$$\% \eta = \frac{\text{Power input} - \text{Power losses}}{\text{Power input}} \times 100\%$$

$$= \frac{1 - \text{Power losses}}{\text{Power input}} \times 100\%$$

(38)

The losses are the active power losses along the line i.e.

$$I^2 R \Rightarrow \text{1-Phase System}$$

$$3I^2 R \Rightarrow \text{3-Phase System}$$

Where:

I = Per phase current

R = Per phase resistance of the line.

Voltage Regulation of a transmission line is defined as the change in voltage at the receiving end expressed as a percentage of a full load voltage at a specified power factor i.e.

$$\text{The Percentage voltage regulation} = \frac{V_{no} - V_{R1}}{V_{R1}} \times 100\%$$

Where: V_{no} = Magnitude of the No-load receiving end voltage

V_{R1} = Magnitude of the Full-load receiving end voltage at a specified power factor.

for a short transmission line.

$$V_{no} \approx V_s \text{ (Sending-end Voltage)}$$

$$V_R \approx V_R \text{ (receiving-end voltage)}$$

$$\therefore \text{Percentage Voltage Regulation} = \frac{V_s - V_R}{V_R} \times 100\%$$

$$= \frac{V_{drop}}{V_R} \times 100\%$$

$$= I(R + jX)$$

$$V_R$$

$$= I [R \cos \phi + j X \sin \phi] \times 100\%$$

(39)

Where: ϕ_r is the angle between the receiving end Voltage and current i.e. the phase angle.

① $\cos \phi_r$ is the receiving-end power factor or the load power factor.

$$\% V.R = \frac{|I| R \cos \phi_r + |I|^2 X \sin \phi_r}{|V_r|} \times 100 \quad \text{--- (6)}$$

In above (6) the phase ϕ_r is positive for lagging load bad - it will be negative for a leading load. Such that;

$$\% V.R = \frac{|I| R \cos \phi_r - |I|^2 X \sin \phi_r}{|V_r|} \times 100 \quad \text{--- (7)}$$

The voltage regulation becomes negative, hence load voltage is more than no-load voltage ($V_r > V_{r_0}$) From Equation 7;

$$X \sin \phi_r > R \cos \phi_r \Rightarrow \frac{\sin \phi_r}{\cos \phi_r} > \frac{R}{X}$$

$$\therefore \tan \phi_r > \frac{R}{X} \quad (\text{Leading load})$$

Similarly, from Equation (8) zero voltage regulation occurs when $X \sin \phi_r = R \cos \phi_r$ and $\tan \phi_r = \frac{R}{X}$

SHORT TRANSMISSION LINE

Transmission lines are classified as short, medium or long transmission line. A short line has a length of 100 km or less. The shunt admittance (jwC) and the line conductance (leakage current) are small and neglected resulting in the simple equivalent circuit Figure 8.

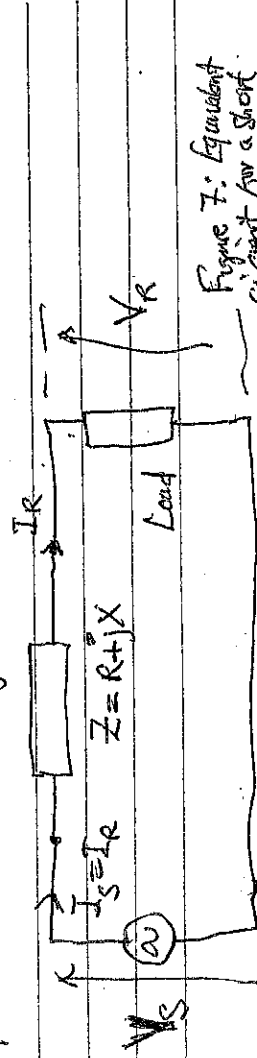


Figure 7: Equivalent circuit for a short line

(40)

This is a Simple Series circuit, the relationship between Sending-end and receiving-end Voltages and currents can be written as follows:

$$V_s = V_r + \text{line (voltage) drop}$$

$$= V_r + I(R + jX_L) = V_r + IR + jIX_L$$

~~$$|V_s| = |V_r| + |I| [R \cos \phi + jX_L \sin \phi]$$~~
~~$$= |V_r| + |I| R \cos \phi + |I| X_L \sin \phi$$~~

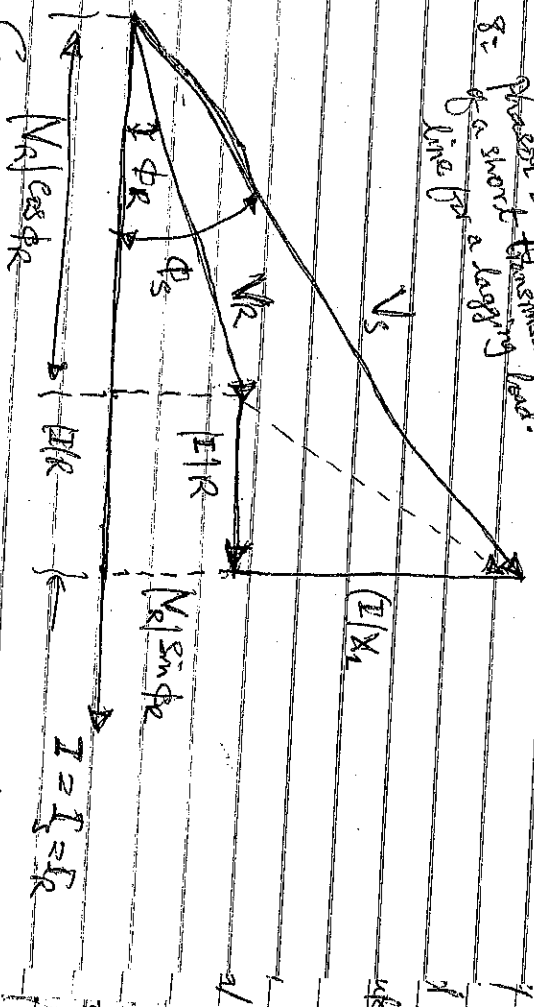
$$V_s = V_r + I [R \cos \phi + jX_L \sin \phi]$$

$$\therefore |V_s| = |V_r| + |I| [R \cos \phi + X_L \sin \phi]$$

$$= |V_r| + |I| R \cos \phi + |I| X_L \sin \phi \quad \text{--- (9)}$$

The phasor diagram for the short transmission line for a lagging load is shown in Figure 8.

Figure 8: Phasor diagram for a short transmission line for a lagging load.



$$V_s^2 = (V_r \cos \phi_r + IR)^2 + (V_r \sin \phi_r + IX_L)^2$$

$$V_s^2 = (V_r \cos \phi_r + IR)^2 + (V_r \sin \phi_r + IX_L)^2$$

$$V_s = |V_r| + |I| (R \cos \phi_r + X_L \sin \phi_r) \quad \text{--- (10)}$$

(41)

Equation (10) gives an expression for the magnitude of the Sending-end Voltage in terms of the load current (I) and the line constants (line inductive reactance (X) and Resistance (R)) under normal load range.

Example: A single-phase 50 Hz generator supplies an inductive load of 5000 kW at a power factor of 0.707 lagging by means of an overhead transmission line 20 km long. The resistance and inductance are 0.0195 Ω and 0.63 mH per kilometre. The voltage at the receiving end is required to be kept at 10 kV. Find the sending-end voltage and voltage regulation of the line. Compute the efficiency of the line.

Solution: The line constants are:

$$R = 0.0195 \Omega \times 20 \text{ km} = 0.39 \Omega$$
$$X_L = 2\pi fL = 2 \times \frac{22}{7} \times 50 \times 0.63 \times 10^{-3}$$

$$= 3.96 \Omega$$

This is short transmission line with $R \neq 0$ and given by

$$P = VI \cos \phi \quad (I) = \frac{P}{V \cos \phi} = \frac{5000}{V \times 0.707}$$

$$(I) = \frac{P}{V \cos \phi} = \frac{5000}{10 \times 0.707} = 707 \text{ A}$$

From Equation (10)

$$|V_s| = |V_r| + |I| R \cos \phi + |I| X \sin \phi$$
$$= 10 + 0.707 \left(\frac{0.39 \times 0.707 + 3.96 \times 0.707}{0.707} \right)$$
$$|V_s| = 12.174 \text{ kV}$$

(42)

$$\text{Voltage Regulation (VR)} = \frac{|V_s| - |V_r|}{|V_r|} \times 100\%$$

$$= \frac{12.75 - 10}{10} \times 100\% = \frac{2.75}{10} \times 100\%$$

$$= \frac{2.75}{10} \times 100\% = 27.5\%$$

(b). Line Efficiency:

$$\eta = \frac{\text{output Power}}{\text{output Power} + \text{Losses}} = \frac{\text{o/p Power}}{\text{o/p Power} + \text{Real Power Losses}}$$

$$\eta = \frac{5000}{5000 + 707^2 \times 0.39 \times 10^{-3}} \times 100\%$$

$$= 96.25\%$$

Example 2: A single-phase transmission line is transmitting

1.1 MW power to a factory at 11 kV and 0.8

power factor lagging. It has a total line resistance

of 2 Ω and reactance of 3 Ω . Determine:

(a) the voltage at the sending end

(b) sending end power factor

(c) Percentage Voltage Regulation

(d) Percentage line efficiency.

Solution:

Note: Unless otherwise specified, all data given is taken as receiving end data by default.

$$P_r = 1.1 \text{ MW} = 1.1 \times 10^6 \text{ W}$$

$$V_r = 11 \text{ kV} = 11 \times 10^3 \text{ V}$$

(43)

$$R = 2 \Omega / \text{phase} \quad X_L = 3 \Omega / \text{phase}$$

The power factor $\cos \phi_R = 0.8$ lagging
 phase angle $\phi_R = \cos^{-1} 0.8 = 36.8^\circ$

$$\therefore \cos \phi_R = 0.8 \quad \sin \phi_R = \sin 36.8^\circ = 0.6$$

$$\text{a) } |V_s| = |V_R| + |I| (R \cos \phi_R + X \sin \phi_R)$$

$$|I| = \frac{P}{|V| \cos \phi_R} = \frac{1.1 \times 10^6}{11 \times 10^3 \times 0.8} = 125 \text{ A}$$

$$\therefore |V_s| = 11 \times 10^3 + 125 (2 \times 0.8 + 3 \times 0.6) =$$

$$= 11425 \text{ V}$$

$$= 11.43 \text{ kV}$$

~~$$\text{b) Sending-end power factor } \phi_s = \phi_R + \delta \quad (\text{where } \delta = \text{phase difference or impedance angle})$$~~

~~$$\text{The impedance angle } \theta \Rightarrow Z \angle \theta = 2 + j3 = 3.6 \angle 56.3^\circ$$~~

~~$$\therefore Z \angle \theta = 3.6 \angle 56.3^\circ ; \theta = 56.3^\circ$$~~

~~$$\phi_s = \phi_R + \theta = 36.8^\circ + 56.3^\circ$$~~

(b) Sending-end power factor ($\cos \phi_s$) where $\phi_s = \phi_R + \delta$

$$\phi_s = \phi_R + \delta \quad ; \quad \delta = \text{Voltage angle}$$

$$V_s = V_R + I \angle \phi_R \times Z \angle \phi = 11000 + 125 \angle 36.8^\circ + 3 \angle 56.3^\circ$$

$$= 11000 + 125 \angle -36.8^\circ + 3.6 \angle 56.3^\circ$$

$$= 11.43 \angle 0.75^\circ = |V_s| \angle \delta$$

$$\therefore \phi_s = \phi_R + \delta = 36.8^\circ + 0.75^\circ = 37.55^\circ$$

(c) efficiency = $\frac{\text{o/p power}}{\text{c/p power + losses}}$ $\times 100\%$ $= \frac{1100000}{1100000 + 31250}$

$$= 97\%$$

$$\text{loss} = 31250 \text{ W}$$

$$= 125 \times 2$$

$$= 31250 \text{ W}$$

(169)

Percentage Voltage Regulation (V_R) = $\frac{(V_S - V_R)}{V_R} \times 100\%$

$$V.R. = \frac{11430 - 11000}{11000} \times 100\%$$

$$= \frac{430}{11000}$$

$$= 0.0391$$

$$= 3.91\%$$

Advantages of P.U. System in Power System analysis :-

- ① The P.U. System gives a clear idea of relative magnitudes of all the power system quantities such, Power, Voltage, Current and Impedance.
- ② The per-unit impedance of equipment of the same rating type have same value regardless of the impedance of the equipment.
- ③ The per unit impedance, voltage and current parameters are same regardless of whether they are referred to the primary or the secondary side.
- ④ The per-unit system are ideal for the computerised analysis and simulation of complex system problems.
- ⑤ The circuit laws are valid in per unit system and the power and voltage phase and line quantities of power and voltage are the same since the factors of $\sqrt{3}$ and 3 are eliminated in per-unit system.

$$V_L = \frac{V}{\sqrt{3}} \text{ phase}$$

$$-(3-\phi) \text{ Power} = 3 \times (1-\phi) \text{ Power}$$

$$S_{3\phi} = 3 V_{\phi} I_{\phi}$$

POWER CABLES

(145)

In big towns and cities and densely populated areas overhead lines become impractical due to safety regulations. In such places conductors are laid underground and are called cables. Though cables are more expensive than overhead lines but are more physically and economically attractive since they can carry large amount of power. The major constraints in the application of cables is the temperature rise of the insulating material used. Power cables consists of three main components:

- (i) Conductor (Core)
- (ii) Insulation (Dielectric)
- (iii) Sheath.

Conductor - The conductor provides the path for current to flow.

The Dielectric - The dielectric is made of insulating

material and serves to withstand the service

voltage and isolate the conductor (core) from

other materials of the cable.

Sheath - The sheath is provided over the cable

insulator to prevent moisture content

into the cable insulation and prevent

the cable from other chemical forces.

It also provides mechanical support to the cable.

DIELECTRIC STRESS IN CABLES

Under normal operating conditions the insulation resistance

of a cable is subjected to electrostatic forces known

as the dielectric stress (E) or electric field

strength / intensity given by:

$$E = \frac{\text{Applied Voltage (V)}}{\text{Thickness of the dielectric}} = \frac{V}{d}$$

In cables this electrostatic field is not uniform,

therefore the dielectric stress (E) at any point in

in a single core cable can change with respect to

distance x . Where x is the internal radius of the

dielectric as shown in Figure 8 below.

(46)

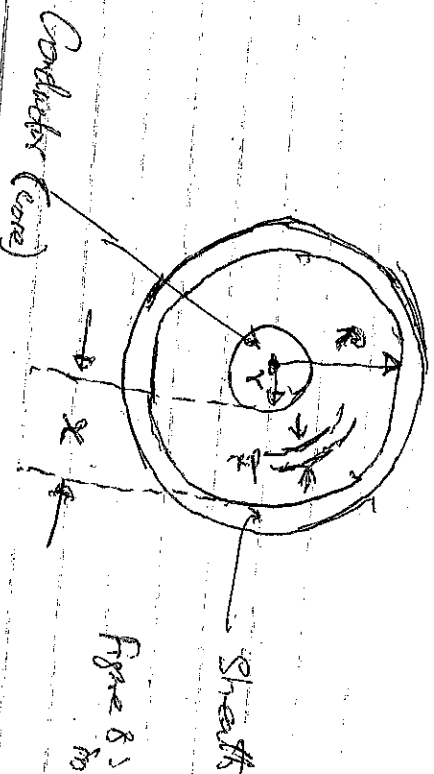


Figure 8.1: Electric stress in a single-core cable

In a single-core cable shown in Figure 8.1, the conductor is surrounded by the dielectric material with an outer metallic sheath. The electric field is confined to the space between the conductor and the sheath and has circular symmetry.

- x = Internal radius of the dielectric
- y = radius of the conductor
- R = radius of sheath (outer radius of insulation)
- ϵ = Permittivity of the dielectric
- ρ = Electric charge in Coulombs/metre of axial length
- V = Voltage between phase and ground
- dx = thickness of annulus

The electric flux density (D) at the distance x from the centre of the conductor is given as:

$$D_x = \frac{Q_x}{S_{Dx}} = \frac{Q_x}{2\pi x l} \quad (1)$$

The dielectric field strength or electric stress is the dielectric material is obtained by varying the distance x from $(x$ to $R)$. The potential difference between the inner radius of the conductor and the outer radius (also known as the potential gradient of the dielectric material) is given as:

$$E_x = \frac{V}{dx} \Rightarrow dv = E_x dx$$

But from $E_x = \frac{D_x}{\epsilon} = \frac{D_x}{\epsilon_0 \epsilon_r}$ (1)

47

$$F_r = \frac{Q}{2\pi\epsilon_0\epsilon_r x} \quad \text{--- (1)}$$

Since $dV = F_r dr$; from (1) $dV = \frac{Q}{2\pi\epsilon_0\epsilon_r} \frac{1}{x} dx$

Integrate both sides; (from inner to the outer radius)

$$\int dV = \int_r^R \frac{Q}{2\pi\epsilon_0\epsilon_r} \frac{1}{x} dx$$

$$V = \frac{Q}{2\pi\epsilon_0\epsilon_r} \ln\left(\frac{R}{r}\right) \quad \text{--- (2)}$$

$$V = \frac{Q}{2\pi\epsilon_0\epsilon_r} (\ln R - \ln r) = \frac{Q}{2\pi\epsilon_0\epsilon_r} \ln\left(\frac{R}{r}\right)$$

$$V = \frac{Q}{2\pi\epsilon_0\epsilon_r} \ln\left(\frac{R}{r}\right) \quad \text{--- (3)}$$

$$\text{From (2)} \quad xF_x = \frac{Q}{2\pi\epsilon_0\epsilon_r} \quad \text{--- (4)}$$

Similarly from (3)

$$V = xF_x \ln\left(\frac{R}{r}\right) \quad \text{--- (5)}$$

∴ The electric field stress in the electric medium between the inner and outer radius and a result of the applied in the conductor can be written from (5) above as:

$$F_x = \frac{V}{x \ln\left(\frac{R}{r}\right)} \quad \text{--- (6)}$$

Thus the electric stress in the cable varies within the insulator (dielectric) and is maximum at the surface of the conductor minimum i.e. $x = R$.

(48)

At $x=R$; $E = E_{(min)}$ and at $x=r$; $E = E_{(max)}$

$$E_{(min)} = \frac{V}{R \ln\left(\frac{R}{r}\right)} \quad ; \quad E_{(max)} = \frac{V}{r \ln\left(\frac{R}{r}\right)}$$

(7)

Example

A single-core cable with a working voltage of 6.5 kV and a conductor diameter of 10 mm. If the dielectric thickness is 7.5 mm, calculate the maximum and minimum electric stress in the insulation.

Phase Voltage $\geq 6.5 \text{ kV} \geq 6.5 \times 1000 \text{ V}$
Inner radius of conductor $= \frac{10 \text{ mm}}{2} = 5 \text{ mm}$

Outer radius of insulation $= R = 5 + 7.5 = 12.5 \text{ mm}$

$$E_{(max)} = \frac{V}{r \ln\left(\frac{R}{r}\right)} = \frac{6.5 \times 1000}{5 \ln\left(\frac{12.5}{5}\right)}$$

$$= 1418.76 \text{ V} = \underline{1.419 \text{ kV/mm}}$$

$$E_{(min)} = \frac{V}{R \ln\left(\frac{R}{r}\right)} = \frac{6.5 \times 1000}{12.5 \ln\left(\frac{12.5}{5}\right)} = 567.505 \text{ V/mm}$$

$$\geq 0.567 \text{ kV/mm}$$

Economical

SPACING OF CABLES

In order to keep a fixed overall size of the cable (R) for a given phase voltage (V), there is a particular conductor size (r) which minimises the electric stress (E_x) i.e. to maximise $\ln\left(\frac{R}{r}\right)$ in order to minimise (E_x). The condition for minimum electric stress (E_x) is obtained by differentiating (E_x) and equating to zero.

(49)

$$\frac{d}{dx} u = V^2 + u^2$$

Quotient Rule.

$$\frac{d}{dx} \left[r \ln \left(\frac{R}{r} \right) \right] = 0 \Rightarrow \frac{d}{dx} [r \ln R - r \ln r] = 0$$

$$= \frac{d}{dx} [r \ln R - r \ln r] = 0$$

$$= \left[\ln R \cdot \frac{d}{dx} (r) + (r) \frac{d}{dx} (\ln R) \right] - \left[\ln r \cdot \frac{d}{dx} (r) + r \frac{d}{dx} (\ln r) \right]$$

$$= (\ln R + 0) - (1 + \ln r) = 0$$

$$= \ln R - 1 - \ln r = 0$$

$$= \ln R - \ln r = 1 \Rightarrow \ln \left(\frac{R}{r} \right) = 1$$

$$\text{When } \ln \left(\frac{R}{r} \right) = 1 \Rightarrow \log \left(\frac{R}{r} \right) = 1$$

$$\therefore \frac{R}{r} = e^1 = 2.71882$$

Hence the maximum $r \ln \left(\frac{R}{r} \right)$ occurs

$$\text{when } R/r = 2.71882$$

$$\frac{r_{\max}}{r_{\min}} = \frac{R}{r}$$

Example 2: A single core cable is operating on three-phase 275 kV supply network. If the maximum dielectric stress (E_x) should not exceed 15 kV/mm. Calculate the overall diameter of the core and insulation (R) and the most economical diameter of the cable core.

Solution:

$$\text{Rms Value of phase Voltage} = \frac{V}{\sqrt{3}}$$

$$= \frac{275}{\sqrt{3}} = 158.814$$

(50)

$$V_{\text{phase}} = \underline{158.8 \text{ kV}}$$

Maximum Value of the phase Voltage $V_{\text{max}} = \sqrt{3} \times V_{\text{phs}}$

$$V_{\text{ph}} = \sqrt{3} \times 158.8 \text{ kV} \\ = \underline{274.54 \text{ kV}}$$

Minimum electric stress / Potential gradient (E_{max}) =

$$E_{\text{max}} = \frac{V_{\text{max}}}{r \ln \left(\frac{R}{r} \right)} = \underline{224.5}$$

For most economical diameter of cable and

case:

$$\ln \left(\frac{R}{r} \right) = 1 \Rightarrow \frac{R}{r} = e^1 = e^1 = 2.7188$$

$$\therefore \frac{R}{r} = 224.5 \Rightarrow 15 = 224.5$$

$$\text{Maximum Case (K)} = \frac{224.5}{15} = \underline{14.96 \text{ mm}}$$

Overall diameter of cable core = $2 \times r$

$$= 2 \times 14.96$$

$$= \underline{29.92 \text{ mm}}$$

$$\text{Since } \frac{R}{r} = 2.718 \Rightarrow R = 14.96 \times 2.718$$

$$= 40.66 \text{ mm}$$

$$\text{Overall dia. of cable} = 2 \times R =$$

$$= 2 \times 40.66$$

$$= \underline{81.32 \text{ mm}}$$

ADVANTAGES OF UNDERGROUND CABLES OVER OVERHEAD LINES

Advantages:-

- ① Less liable to damage due to storm and lightning

(5)

- (2) Low Maintenance cost
- (3) less prone to faults.
- (4) Smaller voltage drop
- (5) better physical appearance

Disadvantages

- (1) Greater Insulation Cost.
- (2) Insulation problems at high voltage.
- (3) Higher cost per unit length.
- (4) Tedious fault location procedure.
- (5) less flexible compared to overhead lines (OTL).

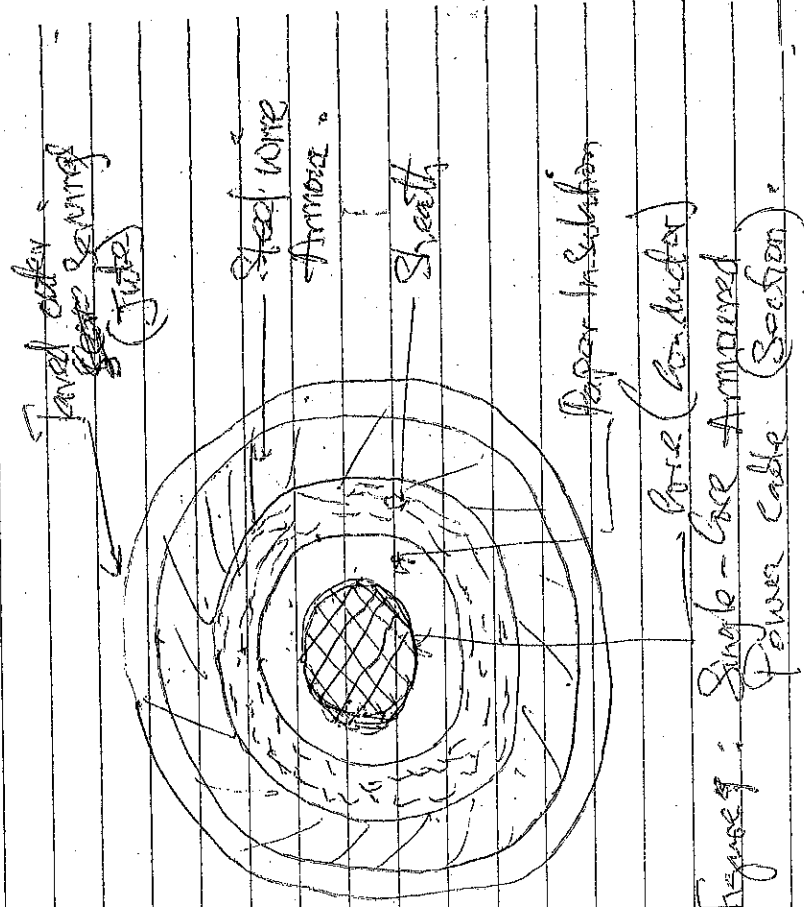


Figure 9: Single-Core Armored Power Cable (Section)